
UNIT 20 ARBITRAGE PRICING THEORY

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20.0 OBJECTIVES

After going through this unit, you will be able to:

- explain the limitations of the CAPM Model;
- describe Ross's critique and extension of the CAPM model;
- describe assumptions of the APT; and
- discuss its empirical study extensions.

20.1 INTRODUCTION

In the intricate realm of financial markets, theories and models serve as essential tools for unravelling the complexities of asset pricing. This Unit embarks on a journey to explore, critique, and extend key asset pricing models, focusing primarily on the Capital Asset Pricing Model (CAPM) and the Arbitrage Pricing Theory (APT).

As we know from Unit 19, the CAPM, born in the 1960s through the collaborative efforts of William Sharpe, John Lintner, and Jan Mossin, laid a foundational understanding of expected returns about systematic risk. However, as financial markets evolved, critiques arose, with Stephen A. Ross leading the charge. Ross's critique of the CAPM centred on its reliance on a single market factor to explain asset prices. In his influential work, "The Arbitrage Theory of Capital Asset Pricing," Ross proposed an alternative approach, suggesting that multiple risk factors impact asset prices. This critique prompted a paradigm shift, urging scholars to move beyond the simplicity of the CAPM towards more nuanced models that could better capture the intricacies of financial markets. We have already discussed about CAPM model in the previous unit, let us discuss its criticism and the APT model as an alternative to the CAPM model.

20.2 ROSS'S CRITIQUE OF THE CAPITAL ASSET PRICING MODEL (CAPM)

Stephen A. Ross, who developed the Arbitrage Pricing Theory (APT), pointed out several drawbacks and simplifying assumptions of the Capital Asset Pricing Model (CAPM). Let us briefly discuss the drawbacks and simplifying assumptions of CAPM model once again.

20.2.1 Drawbacks of CAPM

- 1) **Single-Factor Model:** CAPM is a single-factor model that relies heavily on the market risk factor (beta). It assumes that all systematic risk can be captured by a single factor, which might oversimplify the real-world complexities of asset pricing.
- 2) **Homogeneous Expectations:** CAPM assumes that all investors share the same expectations about future returns, standard deviations, and correlations. This assumption overlooks the diversity of investor beliefs and expectations in the market.
- 3) **Static Risk:** CAPM assumes that the risk associated with an asset is constant over time. In reality, the risk profile of an asset may change due to various factors such as economic conditions, technological advancements, or changes in market sentiment.
- 4) **Risk-Free Rate Assumption:** The model relies on a risk-free rate as a baseline. However, the actual risk-free rate might not be constant, and using government bond yields as a proxy for the risk-free rate assumes that there is no default risk, which may not always hold.

20.2.2 Simplifying Assumptions of CAPM

- 1) **Perfect Markets:** CAPM assumes the existence of perfect and frictionless financial markets. This implies that there are no transaction costs, no taxes, and no restrictions on short selling. In reality, markets are not always perfect, and these assumptions might not hold.

- 2) **Homogeneous Assets:** CAPM assumes that all assets are homogeneous and can be completely described by their expected return and beta. In reality, assets may have unique characteristics that are not captured by these simplified parameters.
- 3) **Normal Distribution of Returns:** The model assumes that asset returns are normally distributed. However, in practice, financial markets often exhibit non-normalities, such as fat tails and volatility clustering, which are not accounted for in the CAPM.
- 4) **Constant Correlations:** CAPM assumes constant correlations between assets. In reality, correlations can change over time, especially during periods of financial distress or market turbulence.
- 5) **Investor Rationality:** CAPM assumes that investors are rational and make decisions solely based on expected returns and risks. Behavioural finance research suggests that investors may not always behave rationally, and psychological factors can influence decision-making.

Understanding these drawbacks and simplifying assumptions is crucial for recognizing the limitations of CAPM and considering alternative models, such as the Arbitrage Pricing Theory (APT), which seeks to address some of these issues by allowing for multiple factors and a more flexible framework.

Check Your Progress 1

- 1) Discuss Ross's critique of CAPM, highlighting drawbacks.
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- 2) What were some simplifying assumptions of the CAPM model?
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20.3 INTRODUCTION TO ARBITRAGE PRICING THEORY

Arbitrage Pricing Theory (APT) is an asset pricing model that was developed by economist Stephen Ross in the early 1970s as an alternative to the Capital Asset Pricing Model (CAPM). The APT is designed to provide a more flexible framework for explaining the expected returns on assets by allowing for multiple sources of systematic risk. Unlike CAPM, APT doesn't rely on the assumption of a single systematic risk factor (market risk), and it does not specify the nature or number of risk factors in advance.

Stephen Ross introduced APT in his seminal paper titled “The Arbitrage Theory of Capital Asset Pricing” published in 1976. The theory was developed as a response to perceived limitations and simplifying assumptions of CAPM. APT acknowledges that an asset’s expected return is influenced by various macroeconomic factors, and it seeks to model these influences more realistically and dynamically.

20.3.1 Assumptions of Arbitrage Pricing Theory (APT)

Systematic and Unsystematic Risk in APT:

- 1) **Separation of Systematic and Unsystematic Risk:** A key assumption of APT is the separation of systematic and unsystematic risk. Systematic risk is captured by the sensitivity of an asset’s return to various macroeconomic factors (as represented by factor betas in the APT model). Unsystematic risk, also known as specific or idiosyncratic risk, is assumed to be diversifiable and can be eliminated through a well-diversified portfolio. APT assumes that investors can eliminate unsystematic risk through proper portfolio construction. We have discussed about portfolio construction in Unit 18.

Market Efficiency and Rational Expectations:

- 2) **Efficient Markets:** APT assumes that financial markets are efficient. In an efficient market, asset prices fully reflect all available information, and mispricing is quickly corrected through the actions of rational and profit-seeking investors. This assumption is crucial for the functioning of arbitrage opportunities, as any deviation from fair value would be exploited by arbitrageurs.
- 3) **Rational Expectations:** APT assumes that investors have rational expectations, meaning they form their expectations about future asset prices based on all available information. Rational expectations are essential for the correct pricing of assets in an efficient market and for the proper functioning of arbitrage.

Arbitrage Opportunities and Their Impact on Asset Pricing:

- 4) **Absence of Arbitrage Opportunities:** A fundamental assumption in APT is the absence of arbitrage opportunities. If arbitrage opportunities were present, investors could make risk-free profits by exploiting mispricing, leading to market adjustments and undermining the model’s ability to explain asset pricing. The assumption of no arbitrage ensures that asset prices accurately reflect their underlying risks.

Rational Behaviour of Investors in APT:

- 5) **Investor Rationality:** APT assumes that investors are rational decision-makers. Rational behaviour implies that investors make choices that maximize their expected utility, given their information and preferences.

Rationality is crucial for the proper functioning of arbitrage and for ensuring that asset prices align with the expectations implied by the model.

In summary, the assumptions of APT revolve around the separation of systematic and unsystematic risk, the efficiency of financial markets, the rational expectations of investors, and the absence of arbitrage opportunities. These assumptions provide the foundation for the model's ability to explain asset pricing dynamics based on the influence of multiple macroeconomic factors.

20.3.2 Theoretical Foundations of APT

The Arbitrage Pricing Theory (APT) provides a framework for understanding the relationship between risk and expected return in financial markets. It was developed by Stephen Ross in 1976 as an alternative to the Capital Asset Pricing Model (CAPM).

The APT formula for expected asset returns is given by:

$$E(R_i) = R_f + \beta_{i1} \times \lambda_1 + \beta_{i2} \times \lambda_2 + \dots + \beta_{ik} \times \lambda_k + \epsilon_i$$

Where:

- $E(R_i)$ is the expected return on asset i ,
- R_f is the risk-free rate,
- β_{ij} represents the sensitivity of asset i to factor j ,
- λ_j is the risk premium associated with factor j ,
- ϵ_i is the specific or idiosyncratic return component.

No Arbitrage Assumption:

APT operates under the assumption of no arbitrage opportunities in the market. This implies that investors cannot consistently earn risk-free profits by exploiting price discrepancies between assets. The absence of arbitrage opportunities ensures that asset prices are efficient and reflect all available information.

Law of One Price:

APT relies on the law of one price, which states that assets with similar risk exposures should have similar expected returns to prevent arbitrage opportunities. In other words, assets that share common risk factors should be priced in such a way that investors are indifferent between them.

Flexibility and Dynamism:

One of the key advantages of APT is its flexibility and dynamism in pricing assets. Unlike CAPM, which relies on a single systematic risk factor (market risk), APT accommodates multiple systematic risk factors. This allows for a

more nuanced understanding of asset pricing, as different assets may be influenced by different sets of risk factors.

Empirical Identification of Factors:

APT allows for the empirical identification of systematic risk factors based on historical data analysis. By analysing the relationship between asset returns and various macroeconomic, financial, and market variables, researchers can identify the factors that drive asset prices and determine their risk premiums.

Significance and Contribution:

The development of APT was a significant contribution to the field of finance, as it offered an alternative perspective to the traditional CAPM framework. By incorporating multiple systematic risk factors and allowing for empirical testing, APT provides a more comprehensive approach to asset pricing that better reflects the complexities of real-world markets.

In summary, APT provides a rigorous framework for understanding the relationship between risk and expected return in financial markets. Its reliance on the absence of arbitrage opportunities, the law of one price, and the accommodation of multiple risk factors make it a valuable tool for both academics and practitioners in the field of finance.

20.3.3 Key Differences between APT and CAPM

1) Number of Factors:

- **CAPM:** The Capital Asset Pricing Model (CAPM) relies on a single systematic risk factor, which is market risk. It assumes that all systematic risk can be captured by the market portfolio.
- **APT:** In contrast, the Arbitrage Pricing Theory (APT) allows for the consideration of multiple systematic risk factors. It recognizes that assets may be influenced by a variety of economic variables such as interest rates, inflation, GDP growth, and industry-specific factors.

2) Factor Identification:

- **CAPM:** CAPM assumes a specific market factor, represented by the market portfolio. This market factor is usually proxied by a broad-based equity index such as the S&P 500.
- **APT:** APT does not predefine factors. Instead, it allows researchers to empirically identify relevant factors based on statistical analysis of historical data. Factors may vary depending on the market and the specific assets being analysed.

3) Sensitivity Measures:

- **CAPM:** In CAPM, the sensitivity of an asset to systematic risk is measured by beta (β). Beta represents the covariance between the

asset's returns and the market portfolio returns, divided by the variance of the market portfolio returns.

- **APT:** APT uses factor sensitivities (β_{ij}). Each beta (β_{ij}) represents the i th asset's sensitivity to a specific factor (j). These factor sensitivities capture how changes in each factor affect the returns of the asset.

4) Assumption about Investor Behaviour:

- **CAPM:** CAPM assumes that investors are rational and risk-averse. It suggests that investors seek to maximize their utility by balancing risk and return and will invest in portfolios that offer the highest expected return for a given level of risk.
- **APT:** APT does not make explicit assumptions about investor behaviour. Instead, it focuses on the relationships between asset returns and macroeconomic factors. It allows for a more flexible and dynamic approach to asset pricing by accommodating different views on investor behaviour.

In summary, while both APT and CAPM are asset pricing models that relate expected returns to systematic risk, they differ in terms of the number of factors considered, the method of factor identification, the sensitivity measures used, and the assumptions about investor behaviour. APT's flexibility in accommodating multiple factors and empirical factor identification makes it a valuable tool for asset pricing analysis in real-world financial markets.

Check Your Progress 2

- 1) What are the key assumptions of the Arbitrage Pricing Theory (APT) regarding the separation of risk, market efficiency, and rational expectations?
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- 2) Provide the APT formula for expected asset returns and explain the significance of the assumptions related to the absence of arbitrage opportunities and investor rationality.
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- 3) Highlight one major difference between the Arbitrage Pricing Theory (APT) and the Capital Asset Pricing Model (CAPM) in terms of the number of systematic risk factors considered, and briefly discuss the implications of this difference.

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20.4 EMPIRICAL STUDIES ON APT

The Arbitrage Pricing Theory (APT) exhibits practical versatility in various applications, particularly in asset allocation, cost of capital computation, and the evaluation of managed funds. The utilization of APT in asset allocation stems from its connection to the factor structure and mean-variance efficiency. With k factors implying the existence of efficient frontier-spanning assets, constructing a mean-variance efficient portfolio becomes more straightforward when k factors are the payoffs of traded securities. APT simplifies the optimization problem by reducing dimensionality when k is small. However, if the factor structure is incorrect, the resulting portfolio may not be mean-variance efficient, necessitating adjustments based on investors' beliefs, as proposed by Pastor and Stambaugh (2000) using Bayesian methods.

In the context of calculating the cost of capital, APT has been applied by various researchers, such as Elton, Gruber, and Mei (1994) and Bower and Schink (1994). However, the challenge lies in the selection of factors, with different studies yielding disparate results. This lack of factor specification is a significant drawback, contributing to the reluctance of entities like the U.S. Federal Reserve Board to adopt APT for certain purposes.

APT's application in evaluating money managers involves regressing managed funds' returns on factors and comparing intercepts with benchmark securities returns. Notable studies, including those by Busse (1999) and Carhart (1997), have explored this application. However, it is essential to recognize the one-period nature of APT and its limitations when pricing derivatives or trading strategies due to potential arbitrage issues arising from the stochastic discount factor.

The APT's ironic challenge is evident in the need for extensions to the basic model in some applications, violating its core tenet of market arbitrage absence. For instance, evaluating dynamic trading strategies may require alternative models like Black-Scholes, as suggested by Glosten and Jagannathan (1994), highlighting the complexity and limitations in the practical application of APT.

Identification of Challenges and Criticisms from Empirical Studies:

- 1) **Data Limitations:** Some empirical studies face challenges related to data limitations. Identifying and collecting accurate historical data for various macroeconomic factors and asset returns can be complex, and the quality of results may depend on the availability and reliability of the data.

- 2) **Model Misspecification:** Critics argue that APT, like any model, is subject to misspecification. If the chosen factors do not adequately capture the relevant sources of systematic risk, the model may provide inaccurate predictions. Empirical studies have explored the sensitivity of APT to the selection of factors.
- 3) **Time-Varying Betas:** Empirical evidence suggests that factor sensitivities (betas) may not be constant over time, challenging the assumption of a static risk profile in APT. Researchers have explored the implications of time-varying betas on the model's performance.
- 4) **Factor Redundancy:** Some studies have found that identified factors may be correlated or redundant, raising questions about the independence of factors assumed by APT. Redundancy in factors could affect the reliability of the model's predictions.
- 5) **Market Efficiency Concerns:** Empirical studies testing APT often assume market efficiency, but real-world markets may deviate from perfect efficiency. If markets are not fully efficient, it could impact the ability of arbitrage to correct mispricing and, consequently, the validity of APT predictions.

In summary, empirical studies on APT have provided valuable insights into the model's performance in explaining asset pricing. While some findings support the theory, challenges and criticisms related to data limitations, model misspecification, time-varying betas, factor redundancy, and market efficiency concerns have been identified. Researchers continue to refine and extend empirical testing to enhance our understanding of APT's strengths and limitations in capturing the complexities of asset pricing.

20.5 CRITICISMS OF APT

20.5.1 Critical Evaluation of APT's Assumptions

- 1) **Arbitrage Opportunities:** Critics argue that the assumption of no arbitrage opportunities might be too restrictive. In real markets, various frictions, transaction costs, and constraints may limit the ability of arbitrageurs to correct mispricing quickly and efficiently.
- 2) **Market Efficiency:** APT relies on the assumption of efficient markets, where prices fully reflect all available information. In practice, markets may not always be perfectly efficient, leading to concerns about the model's applicability in less-than-perfectly-efficient environments.
- 3) **Factor Independence:** APT assumes that factors influencing asset returns are independent. Critics question the independence of these factors, suggesting that correlations or dependencies may exist, challenging the model's assumption of uncorrelated risk factors.
- 4) **Static Betas:** While APT allows for multiple factors, it often assumes that factor sensitivities (betas) are constant over time. Critics argue that

betas can be dynamic, changing with economic conditions and market dynamics and that the static nature of betas may limit the model's accuracy.

- 5) **Investor Rationality:** APT assumes rational expectations, but behavioural finance research suggests that investors may not always behave rationally. Behavioural biases and sentiment-driven behaviour may influence asset pricing, challenging the assumption of investor rationality.

20.5.2 Comparison of APT with Other Asset Pricing Models

- 1) **Comparison with CAPM:** APT has been compared with the Capital Asset Pricing Model (CAPM). Critics argue that while APT addresses some of the limitations of CAPM, it introduces its own set of assumptions and challenges. The choice between APT and CAPM often depends on the specific characteristics of the data and the underlying assumptions deemed most appropriate.
- 2) **Challenges Compared to the Fama-French Model:** The Fama-French Three-Factor Model, an extension of APT, includes market risk, size (SMB), and value (HML) factors. Some critics argue that the Fama-French model may offer better explanatory power than APT, especially in capturing size and value effects observed in empirical studies.

20.5.3 Identifying Areas of Improvement for APT

- 1) **Dynamic Factor Sensitivities:** Critics suggest that allowing for time-varying factor sensitivities in APT could enhance the model's ability to capture changing risk profiles over different market conditions.
- 2) **Incorporating Behavioural Aspects:** Integrating behavioural finance concepts into APT could address the limitations associated with the assumption of investor rationality. Considering psychological factors and market sentiment may provide a more realistic depiction of asset pricing dynamics.
- 3) **Enhancing Factor Identification:** Improving methods for identifying relevant factors is crucial. Empirical studies and model testing should focus on refining the process of selecting macroeconomic variables that genuinely drive asset returns, avoiding redundancy and ensuring independence.
- 4) **Accounting for Nonlinear Relationships:** APT typically assumes linear relationships between factors and asset returns. Acknowledging the possibility of nonlinear relationships may lead to a more accurate representation of the complexities in pricing dynamics.

In conclusion, while APT has been a valuable tool in asset pricing, it is not without criticisms. Addressing issues related to assumptions, considering alternatives, and exploring areas of improvement could contribute to the